Critical Scientific Review

Europa Ice Shell Convection Model:
Physics, Mathematics, Code, and Research Direction

Pre-Thesis Assessment & Actionable Goals

Generated for Joshua — March 2026

Abstract

This document provides a systematic critical review of the EuropaConvection modelling project ahead of thesis writing. It evaluates the physics model, mathematical framework, Monte Carlo uncertainty quantification, Juno observational constraining, and the central scientific hypothesis: that ocean-driven heat delivery to Europa's equator can explain the observed concentration of chaos terrain at low latitudes. The review identifies **three critical gaps** that must be addressed and proposes **three actionable goals** to close them before the thesis is written.

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The Central Hypothesis

What you set out to show

Europa's surface displays a striking geographic asymmetry: chaos terrain (disrupted ice blocks, melt-through features like Conamara Chaos) clusters at equatorial and tropical latitudes, while polar regions are dominated by organised ridged terrain. This is counter-intuitive because tidal heating is expected to peak at the poles ($\varepsilon_{\text{pole}} \approx 2 \times \varepsilon_{\text{eq}}$; Tobie et al. 2003, Beuthe 2013).

Your hypothesis:

> *Ocean circulation preferentially delivers heat to the equatorial ice–ocean interface (Soderlund et al. 2014), thinning the equatorial ice shell and promoting melt-through, diapirism, and surface disruption — producing the observed chaos terrain distribution.*

What the model actually tests

The model tests whether **varying the local basal heat flux** at equatorial surface conditions produces shell thicknesses that are (a) thinner at the equator than at the poles, and (b) consistent with the Juno MWR conductive-lid constraint of $D_{\text{cond}} = 29 \pm 10$ km. It does this via:

1. A 1D transient thermal solver with stagnant-lid convection parameterisation (Green et al. 2021).
2. Monte Carlo uncertainty propagation over 12+ physical parameters.
3. Bayesian importance reweighting against the Juno observation.
4. A 2D latitude-dependent extension with 37 axisymmetric columns.

Review of the Physics Model

Governing equations

The model solves the 1D transient heat equation with volumetric tidal heating:

$$\rho\, c_p \frac{\partial T}{\partial t} = \nabla \cdot (k \nabla T) + q_{\text{tidal}}(T) \tag{1}$$

discretised with Crank–Nicolson (after Rannacher startup) on 31 radial nodes. Harmonic-mean conductivity averaging at interfaces ensures flux conservation across the conductive–convective transition.

Strength

The numerical scheme is well-chosen: Rannacher startup avoids spurious oscillations from stiff initial conditions, Picard iteration handles the nonlinear $k(T)$ and $q_{\text{tidal}}(T)$ couplings, and the tridiagonal solve is $O(N)$. Auto-switching between spherical and Cartesian geometry at $H = 30$ km is a sensible practical choice.

Material properties

Temperature-dependent thermal conductivity $k(T) = 567/T\ \text{W m}^{-1}\,\text{K}^{-1}$ (Howell 2021) is the **single most important equation in the model**. It creates a factor-of-4 conductivity contrast between polar ($T \sim 50$ K, $k \sim 11$ W/m/K) and equatorial ($T \sim 110$ K, $k \sim 5$ W/m/K) surfaces.

Critical Issue

Your own diagnostic (2026-03-20 surface-temperature-dominance analysis) showed that **surface temperature alone accounts for ∼101% of the pole–equator lid-thickness contrast**. This means the $k(T)$ relationship, not ocean heat transport, is doing essentially all the work. The ocean transport multiplier adds a measurable but **secondary** correction. This finding needs to be front and centre in the thesis, not buried.

Rheology and viscosity

The composite diffusion-creep viscosity follows Howell (2021):

$$\eta = \frac{1}{2}\left[\frac{42\,V_m}{R\,T\,d^2}\left(D_v + \frac{\pi\delta}{d}\,D_b\right)\right]^{-1} \tag{2}$$

with volume and grain-boundary diffusivities D_v, D_b following Arrhenius laws.

Strength

Supporting both Maxwell and Andrade rheologies for tidal dissipation is good practice. The Andrade formulation follows McCarthy et al. (2011) and Renaud & Henning (2018), which is the current community standard for icy satellites.

Warning

The grain-size transition at $T = 230$ K (1.0 mm → 0.1 mm) is a hard step function. In reality, dynamic recrystallisation produces a continuous grain-size profile that depends on strain rate, temperature, and stress history (Barr & McKinnon 2007). This step may introduce an artificial viscosity jump at 230 K.

Stagnant-lid convection parameterisation

The model uses the Green et al. (2021) / Deschamps & Vilella (2021) temperature-profile scanning method:

$$\mathrm{Nu} = C\,\mathrm{Ra}^{\xi}\left(\frac{T_i - T_c}{\Delta T}\right)^{\zeta}, \qquad \mathrm{Ra}_{\mathrm{crit}} = 1000 \tag{3}$$

Strength

Dynamic interface detection (scanning for $T \geq T_c$ in the profile) is physically more defensible than fixed-depth cutoffs. The convection ramp for cold columns prevents numerical discontinuities.

Review of the Monte Carlo Framework

Prior distributions

The 2026 parameter audit is well-documented and literature-grounded:

Parameter	Prior	Range	Sensitivity
d_{grain}	LogNormal(0.6 mm, $\sigma = 0.35$ dex)	[0.05, 3.0] mm	**High**
ε_0	LogNormal(1.2×10^{-5}, $\sigma = 0.3$)	$[2 \times 10^{-6}, 3.4 \times 10^{-5}]$	**High**
μ_{ice}	Normal(3.5 ± 0.5 GPa)	[2, 5] GPa	High
q_{basal}	Uniform	[5, 25] mW/m^2	Medium
T_{surf}	Normal(104 ± 7 K)	[80, 120] K	Medium
Q_v	Normal(59.4 kJ/mol, 5%)	—	**Dominant**

Strength

Replacing the old P_{tidal} sampling with direct q_{basal} is a significant improvement. Fixing $f_{\text{salt}} = 0$ (pure-ice baseline) with salinity as a separate scenario avoids the unconstrained salt–conductivity degeneracy of the old prior.

Sobol sensitivity analysis

The Sobol decomposition reveals the model's structural truth:

Parameter	S_1 (first-order)	S_T (total-order)
Q_v (activation energy)	0.365	**0.557**
d_{grain} (grain size)	0.313	**0.496**
ε_0 (tidal strain)	0.096	0.285
q_{basal} (ocean heat)	0.044	**0.142**
T_{surf} (surface temp)	0.014	0.023

Critical Issue

Rheology dominates the conductive lid. The combined S_T for $Q_v + d_{\text{grain}} = 1.05$, while q_{basal} contributes only $S_T = 0.14$. This means the ocean heat flux — the physical quantity your hypothesis depends on — explains only ∼14% of the variance in D_{cond}. The model's own sensitivity analysis shows that **the observable you compare to Juno (D_{cond}) is largely insensitive to the mechanism you are testing (ocean heat transport)**. This is the most important finding in the project and must be honestly confronted in the thesis.

Review of Juno Constraining

Methodology

Bayesian importance reweighting against $D_{\text{cond}}^{\text{obs}} = 29 \pm 10$ km (Levin et al. 2026) with iterative prior tightening. Converged in a single round for all three ocean scenarios at 35° latitude, with ESS fractions ∼69%.

Strength

Importance reweighting is the right choice for an expensive forward model. The ESS fraction of 69% indicates good prior–likelihood overlap, and the PSIS $\hat{k} \approx 0.45$ confirms stable tail behaviour.

Results

All three ocean transport models produce nearly identical posteriors at 35°:

Scenario	q_{mult}	D_{cond} posterior	ESS/N
Uniform	1.000	22.24 km	69.1%
Soderlund	1.002	22.23 km	69.1%
Lemasquerier	0.998	22.26 km	69.1%

Critical Issue

The Juno data cannot discriminate between ocean transport modes. At 35° latitude, the transport multipliers are all ≈ 1.0, so the three scenarios collapse. This is not a failure of the method but a geometric limitation: a single mid-latitude measurement point has no leverage on equator–pole contrasts. The equatorial Bayesian refit (posterior_summary.csv) confirms this: log Bayes factors between scenarios are all < 0.3, far below the threshold for "substantial" evidence.
The thesis must clearly state: *Juno's mid-latitude observation constrains the global mean shell properties but cannot resolve the latitudinal heat transport pattern.*

What the Juno refit did constrain

The meaningful posterior updates are in rheological parameters:

- d_{grain}: shifted from 0.60 mm (prior) to 0.76 mm (posterior), 90% CI narrowed by 55% (from 1.78 to 0.81 dex). But 0.31–1.96 mm is still nearly an order of magnitude.
- q_{basal}: 90% CI narrowed by 19% (from 20 to 16 mW/m^2 width). Median shifted to 10.3 mW/m^2 from prior centre of 15.

Warning

The posterior d_{grain} range of [0.31, 1.96] mm is still broad enough that it cannot distinguish between the Mitri (2023) estimate of 0.39–0.80 mm and the Ruiz et al. (2007) range of 0.2–2 mm.

Review of the 2D Latitude Model

Architecture

The 2D model couples 37 independent 1D columns via operator splitting: radial heat equation first, then explicit lateral diffusion between columns.

Warning

Lateral diffusion is negligible. The thermal diffusion timescale across Europa's radius is $\tau \sim R^2/\kappa \sim 200$ Gyr, far exceeding the model's 15.8 Myr integration window. The 2D model therefore reduces to 37 independent 1D columns with latitude-varying boundary conditions. This is not wrong, but the "2D" framing may overstate the physics. The lateral coupling that matters is **ocean circulation**, which is imposed as a boundary condition, not solved.

Latitude-dependent physics

The latitude profile includes:

1. Surface temperature: $T_s(\phi) = [(T_{\text{eq}}^4 - T_{\text{floor}}^4)\cos^p\phi + T_{\text{floor}}^4]^{1/4}$, calibrated to Ashkenazy (2019).
2. Tidal strain: $\varepsilon_0(\phi) = \varepsilon_{\text{eq}}\sqrt{1 + c\sin^2\phi}$ (Beuthe 2013 whole-shell pattern).

3. Ocean heat flux: scenario-dependent pattern $q(\phi)$ with three options.

Strength

The surface temperature calibration ($p = 1.25$, RMS = 0.70 K vs seasonal energy balance) is well-validated. The Beuthe (2013) strain patterns with multiple geometries (mantle-core, shell-dominated, non-monotonic) provide good coverage of the tidal dissipation uncertainty.

Honest Assessment: The Hypothesis Gap

Here is the uncomfortable truth that the thesis must confront head-on:

The chain of inference has a missing link

Your hypothesis chain is:

$$\underbrace{\text{Ocean heat}}_{\text{cause}} \longrightarrow \underbrace{\text{Thin equatorial ice}}_{\text{mechanism}} \longrightarrow \underbrace{\text{Chaos terrain}}_{\text{observation}}$$

Your model tests the first arrow ($\longrightarrow$) but **never tests the second**. You show that enhanced equatorial heat flux thins H_{total} from 35 km to 24 km (for 1.5× enhancement), but you do not model:

- What shell thickness triggers melt-through or diapir formation.
- Whether thinner ice at the equator is mechanically weaker.
- Whether the predicted thickness contrast matches the observed chaos distribution.

D_{cond} is the wrong observable for this hypothesis

The Juno MWR measures the conductive lid thickness D_{cond}. But your Sobol analysis shows D_{cond} is rheology-dominated ($S_T = 1.05$ for rheological parameters). Meanwhile, H_{total} — the mechanically relevant quantity for melt-through — *is* sensitive to q_{basal} (varying 23–51 km across transport scenarios). The model predicts a strong signal in H_{total} but Juno only measures D_{cond}, where the signal is weak.

Critical Issue

The observable you have (Juno D_{cond}) is insensitive to the mechanism you care about (ocean heat). The quantity that is sensitive (H_{total}) is not directly observable. This is not a flaw in your work — it is a fundamental limitation of the current observational constraints that the thesis should frame honestly.

Surface temperature dominance

Your own analysis (2026-03-20) showed that surface temperature alone produces the equator–pole lid contrast. Ocean heat transport adds a correction, but the baseline story is:

> *Equatorial ice has a thinner conductive lid because warm equatorial surfaces reduce conductivity ($k = 567/T$), not primarily because the ocean delivers more heat there.*

This is actually a **robust, defensible result**. It just shifts the narrative from "ocean dynamics cause chaos" to "surface thermal physics creates the latitude structure, and ocean heat modulates it."

What You Should Actually Be Doing

Given the model's strengths and the gaps identified above, here are the three areas where your effort will have the highest return before thesis writing.

The question you can answer

You cannot definitively prove that ocean heat causes chaos terrain with this model. But you *can* answer a well-posed, publishable question:

Thesis Question

What is the predicted latitude-dependent ice shell structure on Europa under competing ocean heat transport scenarios, and which scenarios are consistent with the Juno MWR constraint?

This reframes the work from "explaining chaos" to "predicting shell structure" — a question your model is well-designed to answer.

Three Actionable Goals Before Thesis Writing

Goal 1: Produce the definitive pole-to-equator shell profiles

What: Run the 2D latitude model (37 columns, 0°–90°) for $N \geq 500$ MC realisations under all three ocean scenarios (uniform, Soderlund, Lemasquerier), with Andrade rheology and the audited 2026 priors. Produce latitude profiles of:

- $D_{\text{cond}}(\phi)$, $D_{\text{conv}}(\phi)$, $H_{\text{total}}(\phi)$ (median $\pm$ 1σ envelopes)
- Convective fraction and Ra/Nu as functions of latitude
- Pole-to-equator thickness ratio $\Delta H/H_{\text{mean}}$

Why: This is the **core deliverable** of the thesis. It provides the predicted observables that Europa Clipper can test. Currently you have 2D runs at $N = 5$–500 but no systematic comparison figure across all three scenarios at production sample size with uncertainty envelopes.
Deliverable: One publication-quality multi-panel figure showing $H_{\text{total}}(\phi)$ and $D_{\text{cond}}(\phi)$ for three scenarios with uncertainty bands. This is Figure 1 of the thesis.
Time estimate: 2–3 days (MC runs + figure generation).

Goal 2: Honest Juno comparison with model discrimination

What: Apply the Bayesian importance reweighting to the 2D latitude profiles (not just 1D endmembers), using the Juno D_{cond} constraint at 35°. Compute:

- Posterior D_{cond} at 35° for each scenario
- Bayes factors between scenarios (uniform vs Soderlund vs Lemasquerier)
- Posterior predictive distributions for D_{cond} at the equator and pole (predictions for Clipper)

Why: The current Juno constraining is done on 1D mid-latitude columns, which collapse all scenarios to identical posteriors. By using the full 2D profiles, you can ask: "Given that Juno sees ∼29 km at 35°, what does each scenario predict at 0° and 90°?" This is where scenario discrimination appears — not at the observation latitude, but at the prediction latitudes.
Deliverable: A table of scenario-conditional predictions at equator/pole/mid-latitude. A figure showing posterior predictive $D_{\text{cond}}(\phi)$ under each scenario. Bayes factors reported honestly (likely inconclusive, which is itself a result).
Time estimate: 3–4 days (reweighting + figure + interpretation).

Goal 3: Sensitivity decomposition by physical mechanism

What: Decompose the pole-to-equator thickness contrast into contributions from:

1. Surface temperature ($k(T)$ effect alone)
2. Tidal heating latitude pattern ($\varepsilon_0(\phi)$ effect)
3. Ocean heat transport ($q_{\text{basal}}(\phi)$ effect)
4. Interaction terms

Do this by running the 2D model with each effect turned on/off:

- Run A: uniform T_s, uniform ε_0, uniform q (control)
- Run B: latitude-varying T_s only (isolate surface effect)
- Run C: latitude-varying $T_s + \varepsilon_0$ (add tidal)
- Run D: latitude-varying all (full model, per scenario)

Why: Your 2026-03-20 surface-temperature-dominance analysis is the most important finding in the project, but it was done as a one-off diagnostic, not as a systematic decomposition with uncertainty. Elevating this to a proper factor-separation experiment (Stein & Alpert 1993) gives you a defensible, quantitative attribution of what controls Europa's shell structure. This is the intellectual core of the thesis.
Deliverable: A bar chart or waterfall diagram showing the contribution of each physical mechanism to ΔH_{total} (pole − equator) with uncertainty bands. This is the "money figure" — the one that says "surface temperature does X%, tidal does Y%, ocean does Z%."
Time estimate: 3–4 days (4 run sets × 3 scenarios + analysis).

Thesis Narrative Recommendation

Based on this review, the honest thesis narrative is:

1. **Europa's ice shell structure is primarily controlled by rheology and surface temperature**, not by ocean heat transport. The temperature-dependent conductivity $k(T) = 567/T$ creates a factor-of-4 conductivity contrast between poles and equator, which alone produces the observed lid-thickness latitude gradient.

2. **Ocean heat transport modulates but does not dominate the shell structure.** Under Soderlund-type equatorial enhancement, the total shell thickness changes by ∼12 km (equatorial thinning), but the conductive lid — the Juno-observable quantity — changes by <2 km.

3. **The Juno MWR constraint is consistent with all tested ocean scenarios** because the observed quantity (D_{cond}) is rheology-dominated. Discriminating between scenarios requires either (a) equator–pole D_{cond} measurements (Clipper), or (b) total shell thickness observations (radar, gravity).

4. **The chaos terrain connection remains an open question.** The model predicts equatorial thinning consistent with enhanced melt probability, but does not model the melt-through process itself. This is a prediction for future coupled thermomechanical models.

This narrative is strong, honest, and publishable. It doesn't overstate the ocean-heat story but frames it as a *testable prediction* rather than a proven mechanism.

Summary of Critical Issues

#	Issue	Severity	Resolution
1	D_{cond} is rheology-dominated, not ocean-heat-driven	**Critical**	Reframe thesis; use H_{total}
2	Juno cannot discriminate ocean scenarios at 35°	**Critical**	Predict at other latitudes
3	Chaos terrain link untested (no melt-through model)	**Major**	Acknowledge as prediction
4	2D lateral diffusion is negligible ($\tau \sim 200$ Gyr)	**Moderate**	Reframe as 1D×37
5	Grain-size step function at 230 K	Minor	Sensitivity test
6	Only Andrade rheology in production	Minor	Add Maxwell comparison

Summary of Strengths

1. **Rigorous parameter audit** with literature-grounded priors for all 12+ parameters.

2. **Sobol sensitivity analysis** providing quantitative attribution of variance to individual parameters — rare in planetary science thermal modelling.

3. **Bayesian observational constraining** via importance reweighting with proper diagnostic checks (ESS, PSIS $\hat{k}$).

4. **Three competing ocean scenarios** spanning the published range of ocean heat transport patterns.

5. **Surface temperature calibration** validated against seasonal energy balance models.

6. **Reproducible codebase** with audited samplers, version-controlled results, and automated figure generation.

Appendix: Key Equations Reference

Thermal conductivity

$$k(T) = \frac{567}{T} \quad [\mathrm{W\,m^{-1}\,K^{-1}}] \tag{4}$$

Composite viscosity (Howell 2021)

$$\eta = \frac{1}{2}\left[\frac{42\,V_m}{R\,T\,d^2}\left(D_{0v}\,e^{-Q_v/RT} + \frac{\pi\delta}{d}\,D_{0b}\,e^{-Q_b/RT}\right)\right]^{-1} \tag{5}$$

Maxwell tidal heating

$$q_{\mathrm{tidal}} = \frac{\varepsilon_0^2\,\omega^2\,\eta}{2\,(1+\omega^2\eta^2/\mu^2)} \tag{6}$$

Andrade complex compliance

$$J^*(\omega) = J_U + \beta\,\Gamma(1+\alpha)\,(\omega\tau)^{-\alpha}\,e^{-i\alpha\pi/2} - \frac{i}{\omega\eta} \tag{7}$$

Rayleigh number

$$\mathrm{Ra} = \frac{\rho\,g\,\alpha_T\,\Delta T\,D_{\mathrm{conv}}^3}{\kappa\,\eta} \tag{8}$$

Surface temperature profile

$$T_s(\phi) = \left[(T_{\mathrm{eq}}^4 - T_{\mathrm{floor}}^4)\cos^p\phi + T_{\mathrm{floor}}^4\right]^{1/4}, \quad p = 1.25 \tag{9}$$

Tidal strain latitude dependence (Beuthe 2013)

$$\varepsilon_0(\phi) = \varepsilon_{\mathrm{eq}}\sqrt{1 + c\,\sin^2\phi}, \quad c = \left(\frac{\varepsilon_{\mathrm{pole}}}{\varepsilon_{\mathrm{eq}}}\right)^2 - 1 \tag{10}$$